An Offline Metric for the Debiasedness of Click Models

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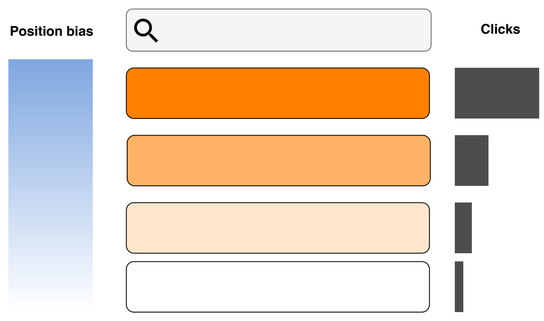
Neural click models

- **Goal**: Predict user interactions with a given ranking of items
- **Model effects that impact clicks**: item relevance, position bias, trust in the system, scroll direction, abandoning a search session...
- **Applications**: Ad click prediction, user understanding, user simulation in RL, bias estimation for unbiased and fair learning-to-rank

Click model applications require click predictions on rankings that are out of the training distribution (ood prediction)

Evaluation protocol fails to predict ood performance

- **Click prediction** using log-likelihood or perplexity only captures in-distribution performance [1]
- **Relevance estimation** using nDCG against human relevance annotations fails when the logging policy is already good [1]



Ignoring position bias and naively interpreting clicks as relevance can achieve high nDCG scores

Current solutions

- Obtain training data from multiple, diverse logging policies
- Simulate ood setting

Can we detect a lack of robustness on a single (observational) dataset?

Comparing model mistakes

Debiasedness: The predicted relevance of an item is not influenced by where the logging policy placed that item.

$$\tilde{R}^{\mathcal{D}} \perp R_l \mid (R, \mathcal{D})$$

$\tilde{R}^{\mathcal{D}}$: Relevance estimations by a click model after training on dataset $\mathcal{D}$

R_l: Relevance estimated by the logging policy

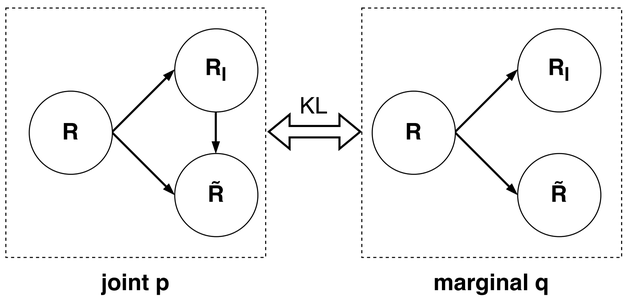
R: Ground-truth relevance scores obtained by human annotators

Quantifying debiasedness as the Conditional Mutual Information w.r.t the logging Policy (CMIP):

$$\tilde{R} \perp R_l \mid R \iff I(\tilde{R}; R_l \mid R) = 0$$

Estimating CMIP

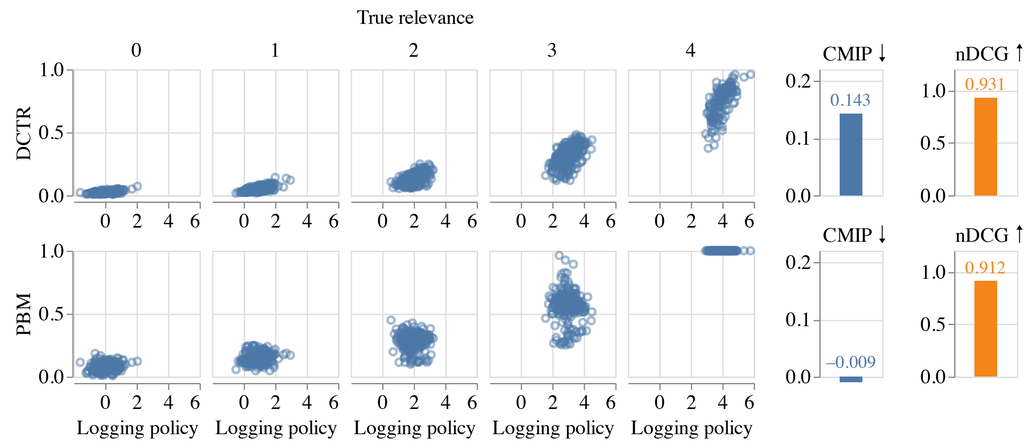
$$\text{CMIP} = I(\tilde{R}; R_l \mid R) = \mathcal{D}_{\text{KL}}(p \| q)$$



Approach:

1. **Simulate samples from marginal distribution** using KNN-based permutations [3]
2. **Estimate KL divergence between joint and marginal distribution** by casting KL-divergence as a classification problem [2]

Visual intuition



Comparing CMIP and nDCG for a PBM and (biased) DCTR click model
The higher CMIP indicates that the DCTR replicates errors of the logging policy

Predicting ood performance in click simulation

Semi-synthetic click simulation

1. Rank items using one of three logging policies
2. Simulate train clicks on rankings using four different user models
3. Train seven different neural click models
4. Simulate test clicks on rankings obtained by a different logging policy
5. Evaluate ood performance of models using perplexity on test clicks

Using metric combinations to predict the out-of-distribution performance of click models using regression, reporting the Adjusted R^2

User model	Logging policy	Test policy	ind PPL	nDCG	CMIP	**ind PPL, nDCG**	CMIP, ind PPL	CMIP, nDCG	Joint
PBM	NoisyOracle	LambdaMART	0.276	0.256	0.439	0.395	$0.938^{\blacktriangle}$	$0.972^{\blacktriangle}$	$0.963^{\blacktriangle}$
		Uniform	0.277	0.346	$0.725^{\blacktriangle}$	0.304	$0.966^{\blacktriangle}$	$0.969^{\blacktriangle}$	$0.970^{\blacktriangle}$
	LambdaMART	NoisyOracle	$0.369^{\blacktriangledown}$	0.907	0.904	0.861	0.883	$0.962^{\blacktriangle}$	$0.944^{\blacktriangle}$
		Uniform	$0.347^{\blacktriangledown}$	$0.965^{\blacktriangle}$	0.931	0.889	$0.974^{\blacktriangle}$	$0.977^{\blacktriangle}$	$0.975^{\blacktriangle}$
DBN	NoisyOracle	LambdaMART	$0.235^{\blacktriangledown}$	$0.978^{\blacktriangle}$	$0.517^{\blacktriangledown}$	0.693	$0.787^{\blacktriangle}$	$0.977^{\blacktriangle}$	$0.897^{\blacktriangle}$
		Uniform	$0.254^{\blacktriangledown}$	$0.984^{\blacktriangle}$	$0.545^{\blacktriangledown}$	0.688	$0.821^{\blacktriangle}$	$0.983^{\blacktriangle}$	$0.963^{\blacktriangle}$
	LambdaMART	NoisyOracle	$0.941^{\blacktriangledown}$	$0.997^{\blacktriangle}$	$0.997^{\blacktriangle}$	0.974	0.970	$0.997^{\blacktriangle}$	0.983
		Uniform	$0.958^{\blacktriangledown}$	$0.999^{\blacktriangle}$	$0.999^{\blacktriangle}$	0.983	0.982	$0.999^{\blacktriangle}$	0.982
MixtureDBN	NoisyOracle	LambdaMART	$0.805^{\blacktriangledown}$	$0.149^{\blacktriangledown}$	0.843	0.869	$0.910^{\blacktriangle}$	$0.906^{\blacktriangle}$	$0.914^{\blacktriangle}$
		Uniform	$0.618^{\blacktriangledown}$	$0.266^{\blacktriangledown}$	$0.859^{\blacktriangle}$	0.767	$0.884^{\blacktriangle}$	$0.909^{\blacktriangle}$	$0.880^{\blacktriangle}$
	LambdaMART	NoisyOracle	0.985	$0.167^{\blacktriangledown}$	$0.625^{\blacktriangledown}$	0.988	0.990	$0.862^{\blacktriangledown}$	0.985
		Uniform	$0.951^{\blacktriangledown}$	$0.167^{\blacktriangledown}$	$0.735^{\blacktriangledown}$	0.987	0.984	$0.902^{\blacktriangledown}$	0.987
Carousel	NoisyOracle	LambdaMART	0.974	$0.811^{\blacktriangledown}$	$0.993^{\blacktriangle}$	0.970	0.977	0.978	0.973
		Uniform	0.954	0.941	$0.994^{\blacktriangle}$	0.947	0.968	$0.995^{\blacktriangle}$	0.942
	LambdaMART	NoisyOracle	0.993	$0.129^{\blacktriangledown}$	$0.688^{\blacktriangledown}$	0.990	0.991	$0.773^{\blacktriangledown}$	0.994
		Uniform	0.993	$0.113^{\blacktriangledown}$	$0.722^{\blacktriangledown}$	0.996	0.994	$0.759^{\blacktriangledown}$	0.998
Average			0.683	0.574	0.782	0.831	0.939	0.933	0.959

Using CMIP in addition to perplexity and nDCG helps to predict the out-of-distribution performance of neural click models

Findings & Limitations

- **CMIP improves the prediction of downstream performance of click models when coupled with existing metrics**
- **CMIP lowers the regret incurred by off-policy selection strategies**

Limitations: CMIP is a pointwise metric, requires relevance annotations, and assumes that these annotations are a perfect predictor of relevance

References

[1] Deffayet, R., Renders, J.-M., and de Rijke, M. Evaluating the robustness of click models to policy distributional shift. In *ACM TOIS 2023*.

[2] Mukherjee, S., Asnani, H., and Kannan, S. CCMI: Classifier-based conditional mutual information estimation. In *UAI 2019*.

[3] Sen, R., Suresh, A. T., Shanmugam, K., Dimakis, A. G., and Shakkettai, S. Model-powered conditional independence test. In *NIPS 2017*.

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